DANA 4830 Final Exam — master overview (Leo Sanchez)

Submission PDF. Entregar `04_Analysis/2_0_final_overview.pdf`, generado desde **este notebook** (`03_Codes/2_0_final_overview.ipynb`). Los otros notebooks son detalle; el profesor puede revisar solo el PDF (o este notebook ejecutado).

Para regenerar solo el entregable: `python run_all.py --overview-pdf` desde `03_Codes/`.

Numbers come from saved CSVs in `04_Analysis` (not typed by hand).

```
Rosati → DGE / biological evidence
        +
Syed → ML / predictive feature selection
        ↓
Candidate biomarkers → evaluation and external comparison
```

Key terms used in this notebook

Term	Simple explanation
Candidate biomarker	A gene with support in this analysis. Not a clinical test yet.
Transferability	Whether a panel from one disease still works in another.
Study-wise centering	Simple batch adjustment. Not ComBat.

Main methodological idea

DGE and machine learning answer related but different questions. DGE finds genes with statistically important expression changes. ML feature selection finds genes that help predict the class. Using both can shorten the candidate list and give two kinds of evidence. Strong results in one dataset are not enough for clinical validation.

PCA is only a picture of variation. A VAE is optional and not used, because latent variables would not be individual gene names.

Task 1.1 — written comparison

Full ~300-word text is embedded below from `04_Analysis/task1_1_comparison.md`.

Short idea: Rosati = which genes change (limma here); Syed/Midterm = which subset predicts class. The two pipelines **complement** each other: DGE narrows the biological search space under FDR control; ML then shrinks to a compact predictive panel.

On GSE75214 we reached a **16-gene ML consensus panel** (all numeric expression features). The LD1 teaching plot shows disease vs control largely separated on that panel, so **LDA with shrinkage** was a natural linear baseline — **mean 5-fold CV ROC-AUC 0.9948** (SMOTE inside each fold).

Dimensionality reduction. PCA in Task 2.2 is exploratory only. A VAE is not used: latent axes are not gene names.

Setup

In [1]:

```
# We only load saved outputs here, so this notebook is fast and cannot drift from t
import pathlib
import pandas as pd
from IPython.display import Image, display

OUT = pathlib.Path("../04_Analysis")
FIG = OUT / "figures"
DATA = pathlib.Path("../01_Data")
pd.set_option("display.width", 120)

def show_fig(path, width=560):
    path = pathlib.Path(path)
    if path.exists():
        display(Image(str(path), width=width))
    else:
        print(f"Figure missing: {path}")

print("Loaded paths. Figures in", FIG)
```

```
Loaded paths. Figures in ..\04_Analysis\figures
```

Notebook map

Notebook	What it does
`2_1_01_data_and_dge.ipynb`	GSE75214 load, skim, limma DGE, volcano, heatmap modules
`2_1_02_feature_selection.ipynb`	MI, RFECV, EN, GBC consensus panel (train-only SMOTE)
`2_1_03_evaluation_and_comparison.ipynb`	LDA/LR/RF CV, WELM/BWELM, enrichment, Midterm compare
`2_2_01_merge_datasets.ipynb`	Four GEO datasets, raw skim, merge, study-wise centering

Notebook	What it does
`2_2_02_validate_pipelines.ipynb`	Cervical DGE, Wang 10/10, ML panel 47 genes
`2_0_final_overview.ipynb`	This master summary + oral defense

In [2]:
```
task11 = (OUT / "task1_1_comparison.md").read_text(encoding="utf-8")
print(task11)
```

```
# Task 1.1 — Comparison of biomarker approaches (~300 words)

Rosati et al. (2024) review a statistics-first bioinformatics pipeline: quality
control, normalization, differential expression, multiple-testing control (FDR),
pathway enrichment, then ROC/AUC and, when follow-up exists, survival models. The
scientific question is *which genes are differentially expressed* and what biology
they point to. Depending on the assay, that DGE step may use edgeR (TMM + negative
binomial), DESeq2 (median-of-ratios + negative binomial), NOISeq, SAMseq, or limma
(linear models with empirical Bayes). GSE75214 is a microarray of log2 intensities,
not raw RNA-seq counts, so limma is the appropriate tool here; swapping in DESeq2
or edgeR would be methodologically wrong for this matrix.

Syed et al. (2024), which Midterm 1 replicated as `syed_pipeline`, ask a different
question: *which small subset best predicts class*? After DEG filtering, four
feature-selection methods (mutual information, RFECV, elastic net, gradient boostin
g)
are combined into a compact panel, class imbalance is treated (SMOTE or RUS plus
weights), and classifiers are scored with CV metrics. Linear PCA can visualize
samples; a VAE is a possible nonlinear reducer but is not used here, because latent
dimensions are no longer named genes and would weaken biomarker interpretability.

The two pipelines complement each other rather than compete. DGE reduces the
biological search space under FDR control and points to pathways; machine learning
then shrinks the predictive feature space to a compact panel. On GSE75214 this exam
found 535 IBD DEGs (FDR < 0.05, |log2FC| ≥ 1.0) and a **16-gene ML consensus panel**
after train-only SMOTE. Because expression values are numeric and the LD1 teaching
plot on that panel shows disease and control samples largely separated, LDA was a
natural linear baseline: shrinkage LDA on the ML panel reached **mean CV ROC-AUC
0.9948** (SMOTE inside each fold). That supports the idea that the selected genes
carry a strong linear signal in this cohort, while logistic regression and random
forest give similar high AUC as extra checks.

None of the eight Midterm 1 breast-cancer Table-3 biomarkers overlapped that IBD
DEG set (three of eight are present on the GSE75214 platform). That does not
invalidate the midterm model; it shows limited cross-disease transfer. Overall, DGE
is useful for statistically stable, biologically interpretable signals; ML is useful
for compact prediction panels. Using both gives a better balance between biological
meaning and predictive performance within the disease studied.
```

In [3]:
```
# Headline metrics computed from saved analysis tables (not typed by hand).
import json
import pandas as pd
from pathlib import Path
from IPython.display import display

dge1 = pd.read_csv(OUT / "dge_table_2_1.csv", index_col=0)
```

```python
deg1 = dge1[dge1["DEG"]]
ml1_n = len(pd.read_csv(OUT / "ml_biomarkers_2_1.csv", index_col=0))
master = pd.read_csv(DATA / "cervical_master.csv")
dge2 = pd.read_csv(OUT / "cervical_dge_table.csv", index_col=0)
strict2 = (dge2["FDR"] < 0.05) & (dge2["log2FC"].abs() >= 1.5)
cdeg2 = dge2[strict2]
ml2_n = len(pd.read_csv(OUT / "cervical_ml_panel.csv", index_col=0))
wang_hub = ["CDK1", "CDC20", "AURKA", "TOP2A", "ASPM", "NCAPG", "KIF23", "CENPF", "
wang_rec = sum(1 for g in wang_hub if g in dge2.index and bool(strict2.loc[g]))

rows = [
    ("GSE75214 DEGs (|log2FC|>=1.0)", int(deg1.shape[0])),
    ("GSE75214 DEGs (|log2FC|>=1.5)", int(((dge1["FDR"] < 0.05) & (dge1["log2FC"].a
    ("GSE75214 ML panel genes", ml1_n),
    ("Master samples", master.shape[0]),
    ("Master common genes", master.shape[1] - 2),
    ("Master DEGs (|log2FC|>=1.5)", int(cdeg2.shape[0])),
    ("Master ML panel genes", ml2_n),
    ("Wang hub genes recovered", f"{wang_rec}/10"),
]
summary = pd.DataFrame(rows, columns=["Metric", "Value"])
display(summary)

if (OUT / "report_data.json").exists():
    report = json.loads((OUT / "report_data.json").read_text(encoding="utf-8"))
    check = pd.DataFrame([
        ("t21_deg", report["t21_deg"], int(deg1.shape[0])),
        ("t22_deg", report["t22_deg"], int(cdeg2.shape[0])),
        ("t22_wang_rec", report["t22_wang_rec"], wang_rec),
    ], columns=["Key", "report_data.json", "From CSVs"])
    print("Sanity check vs report_data.json:")
    display(check)
```

	Metric	Value
0	GSE75214 DEGs (|log2FC|>=1.0)	535
1	GSE75214 DEGs (|log2FC|>=1.5)	158
2	GSE75214 ML panel genes	16
3	Master samples	179
4	Master common genes	12191
5	Master DEGs (|log2FC|>=1.5)	381
6	Master ML panel genes	47
7	Wang hub genes recovered	10/10

```
Sanity check vs report_data.json:
```

	Key	report_data.json	From CSVs
0	t21_deg	535	535
1	t22_deg	381	381
2	t22_wang_rec	10	10

Dataset dimensions and computational reduction

In [4]:

```
# Numbers come from saved CSV outputs shown in the notebooks below.
from IPython.display import display

audit_p = OUT / "cervical_dataset_audit.csv"
dge1 = pd.read_csv(OUT / "dge_table_2_1.csv", index_col=0)
ml1 = pd.read_csv(OUT / "ml_biomarkers_2_1.csv", index_col=0)
dge2 = pd.read_csv(OUT / "cervical_dge_table.csv", index_col=0)
ml2 = pd.read_csv(OUT / "cervical_ml_panel.csv", index_col=0)
master = pd.read_csv(DATA / "cervical_master.csv")
gse = pd.read_csv(DATA / "gse75214_clean.csv", index_col=0)
cmp1 = pd.read_csv(OUT / "panel_comparison_2_1.csv")

rows = [
    {"Task": "2.1", "Dataset": "GSE75214",
     "Samples": gse.shape[0], "Features/Genes": gse.shape[1] - 1,
     "Target": "IBD / Control",
     "Class split": f"{int((gse['target']==1).sum())} / {int((gse['target']==0).sum
]
if audit_p.exists():
    aud = pd.read_csv(audit_p)
    print("Cervical dataset audit (full table):")
    display(aud)
    for _, r in aud.iterrows():
        rows.append({"Task": "2.2", "Dataset": r["Dataset"],
                     "Samples": int(r["Retained samples"]),
                     "Features/Genes": int(r["Genes after mapping"]),
                     "Target": "Cancer / Normal",
                     "Class split": f"{int(r['Cancer'])} / {int(r['Normal'])}"})
rows.append({"Task": "2.2", "Dataset": "Master (shared genes)",
             "Samples": master.shape[0],
             "Features/Genes": master.shape[1] - 2,
             "Target": "Cancer / Normal",
             "Class split": f"{int((master['target']==1).sum())} / {int((master['ta
dim_tbl = pd.DataFrame(rows)
display(dim_tbl)

red21 = pd.DataFrame([
    {"Stage": "Raw GSE75214 probes", "Input": gse.shape[1] - 1, "Output": gse.shape
    {"Stage": "Significant DEGs (FDR<0.05, |log2FC|>=1)", "Input": int((~dge1['DEG'
     "Output": int(dge1['DEG'].sum())},
    {"Stage": "Strong DEG subset (|log2FC|>=1.5)", "Input": int(dge1['DEG'].sum()),
     "Output": int(((dge1['DEG']) & (dge1['log2FC'].abs() >= 1.5)).sum())},
    {"Stage": "ML candidate panel", "Input": int(dge1['DEG'].sum()), "Output": len(
])
```

```python
red22 = pd.DataFrame([
    {"Stage": "Four mapped matrices (shared-gene step)", "Input": "platform-specifi
     "Output": master.shape[1] - 2},
    {"Stage": "DGE on master", "Input": master.shape[1] - 2, "Output": int(dge2['DE
    {"Stage": "ML panel on cervical DEGs", "Input": int(dge2['DEG'].sum()), "Output
])
print("Task 2.1 reduction")
display(red21)
print("Task 2.2 reduction")
display(red22)

print("Task 2.1 ML panel (full):")
display(ml1.round(4))
print("Task 2.1 panel comparison:")
display(cmp1)
print("Task 2.2 ML panel (full):")
display(ml2.round(4))
```

```
Cervical dataset audit (full table):
```

	Dataset	Raw orientation	Raw samples	Raw probes	Excluded samples	Retained samples	Cancer	Normal	Probes mapped to a symbol	I
0	GSE6791	probes × samples	84	54675	56	28	20	8	45118	
1	GSE9750	probes × samples	66	22283	9	57	33	24	21156	
2	GSE63514	probes × samples	128	54675	76	52	28	24	45118	
3	GSE67522	probes × samples	42	42670	0	42	20	22	28282	

	Task	Dataset	Samples	Features/Genes	Target	Class split
0	2.1	GSE75214	194	33252	IBD / Control	172 / 22
1	2.2	GSE6791	28	22836	Cancer / Normal	20 / 8
2	2.2	GSE9750	57	13909	Cancer / Normal	33 / 24
3	2.2	GSE63514	52	22836	Cancer / Normal	28 / 24
4	2.2	GSE67522	42	19430	Cancer / Normal	20 / 22
5	2.2	Master (shared genes)	179	12191	Cancer / Normal	101 / 78

```
Task 2.1 reduction
```

	Stage	Input	Output
0	Raw GSE75214 probes	33252	33252
1	Significant DEGs (FDR<0.05, |log2FC|>=1)	21147	535
2	Strong DEG subset (|log2FC|>=1.5)	535	158
3	ML candidate panel	535	16

```
Task 2.2 reduction
```

	Stage	Input	Output
0	Four mapped matrices (shared-gene step)	platform-specific	12191
1	DGE on master	12191	381
2	ML panel on cervical DEGs	381	47

```
Task 2.1 ML panel (full):
```

gene	log2FC	FDR	direction	n_methods	MI	RFECV	ElasticNet	GBC
IFITM3	1.2432	0.0000	up	3	True	False	True	True
CNTFR	-1.9839	0.0000	down	3	True	False	True	True
PDZK1IP1	1.6307	0.0000	up	3	True	False	True	True
BACE2	1.3638	0.0000	up	3	True	True	True	False
SLC6A14	4.5331	0.0000	up	3	True	False	True	True
STXBP1	1.0620	0.0000	up	3	False	True	True	True
PAQR5	-1.9605	0.0000	down	2	True	False	False	True
SLC38A4	-1.5309	0.0000	down	2	True	False	True	False
TIMP1	2.1728	0.0000	up	2	True	False	False	True
NAAA	-1.2886	0.0000	down	2	True	False	False	True
TINAG	-1.3247	0.0000	down	2	False	True	True	False
ZG16B	1.0237	0.0000	up	2	False	True	True	False
SLC9A3	-1.0528	0.0002	down	2	False	True	False	True
CD34	1.0010	0.0000	up	2	False	True	True	False
APOBEC3B	-1.2465	0.0000	down	2	False	True	False	True
LOC101060256	1.0533	0.0032	up	2	False	False	True	True

```
Task 2.1 panel comparison:
```

	comparison	shared	out_of
0	Machine learning within statistics panel	6	16
1	Machine learning within Midterm 1 panel	0	16
2	Statistics within Midterm 1 panel	0	8
3	Midterm 1 genes present in GSE75214	3	8

```
Task 2.2 ML panel (full):
```

gene	log2FC	FDR	direction	n_methods
SMC4	1.9213	0.0	up	4
THSD4	-2.6903	0.0	down	4
CDKN2A	3.5719	0.0	up	4
UBE2C	1.7436	0.0	up	4
TACC3	1.7440	0.0	up	4
TYMS	2.0525	0.0	up	3
TIMELESS	1.7836	0.0	up	3
RAD54L	2.2075	0.0	up	3
MCM4	1.6411	0.0	up	3
FANCI	1.8036	0.0	up	3
PIGR	-2.0752	0.0	down	3
TBX3	-2.1246	0.0	down	3
TUBB2A	-1.9423	0.0	down	3
PHYHIP	-2.2783	0.0	down	2
MELK	2.3539	0.0	up	2
BBOX1	-3.6108	0.0	down	2
PRC1	2.0804	0.0	up	2
CRNN	-7.0438	0.0	down	2
STIL	1.7936	0.0	up	2
AURKA	2.3028	0.0	up	2
CDC25B	1.8196	0.0	up	2
CA9	2.4590	0.0	up	2
KNTC1	2.4066	0.0	up	2
SULF1	1.7187	0.0	up	2
CELSR3	1.9964	0.0	up	2
PPP1R3C	-4.1164	0.0	down	2
SLC15A3	1.7077	0.0	up	2
KIFC1	1.7751	0.0	up	2
AIM2	2.8498	0.0	up	2

gene	log2FC	FDR	direction	n_methods
MUC5B	-1.9375	0.0	down	2
LHX2	1.9454	0.0	up	2
SPINK2	-1.8204	0.0	down	2
CYP3A5	-2.1226	0.0	down	2
TFF3	-2.0511	0.0	down	2
MASP1	-1.5950	0.0	down	2
LGALSL	-2.0129	0.0	down	2
CRISP2	-2.4808	0.0	down	2
ZSCAN18	-2.0014	0.0	down	2
ZNF415	-1.7473	0.0	down	2
INHBA	3.0640	0.0	up	2
CXCR2	-2.1338	0.0	down	2
CXCL8	3.5600	0.0	up	2
DPP4	-1.8245	0.0	down	2
LPAR6	-1.5929	0.0	down	2
DUSP1	-1.5484	0.0	down	2
KCNK7	-2.1802	0.0	down	2
SYCP2	3.5861	0.0	up	2

Dataset acquisition checklist

In [5]:
```
# Dataset acquisition checklist (Samuel-style columns, Leo's executed audit data)
audit = pd.read_csv(OUT / "cervical_dataset_audit.csv")
checklist = audit.assign(
    Expression_scale=audit["log2 applied"].map(
        {True: "raw → log2 applied", False: "log-like (no log2)"}
    ),
    Gene_annotation_completed="Yes",
).rename(columns={
    "Dataset": "Dataset",
    "Platform": "Platform",
    "Cancer": "Cancer samples",
    "Normal": "Normal samples",
})[[
    "Dataset", "Platform", "Cancer samples", "Normal samples",
    "Expression_scale", "Gene_annotation_completed",
]]
```

```python
print("Dataset acquisition checklist:")
display(checklist)
```

```
Dataset acquisition checklist:
```

	Dataset	Platform	Cancer samples	Normal samples	Expression_scale	Gene_annotation_completed
0	GSE6791	GPL570	20	8	log-like (no log2)	Yes
1	GSE9750	GPL96	33	24	raw → log2 applied	Yes
2	GSE63514	GPL570	28	24	log-like (no log2)	Yes
3	GSE67522	GPL10558	20	22	raw → log2 applied	Yes

DGE reduces the biological search space. ML feature selection reduces it again to a shorter candidate list. These counts come from the executed tables. Strong results in one dataset are still not a clinical validation.

Task 1.2 The two pipeline diagrams

Diagram 1 is the classical statistics driven pipeline from Rosati et al. (2024). **Diagram 2** is the machine learning pipeline conducted in Midterm 1.

In [6]:
```python
show_fig(FIG / "diagram1_rosati_dge_pipeline.png", width=620)
```

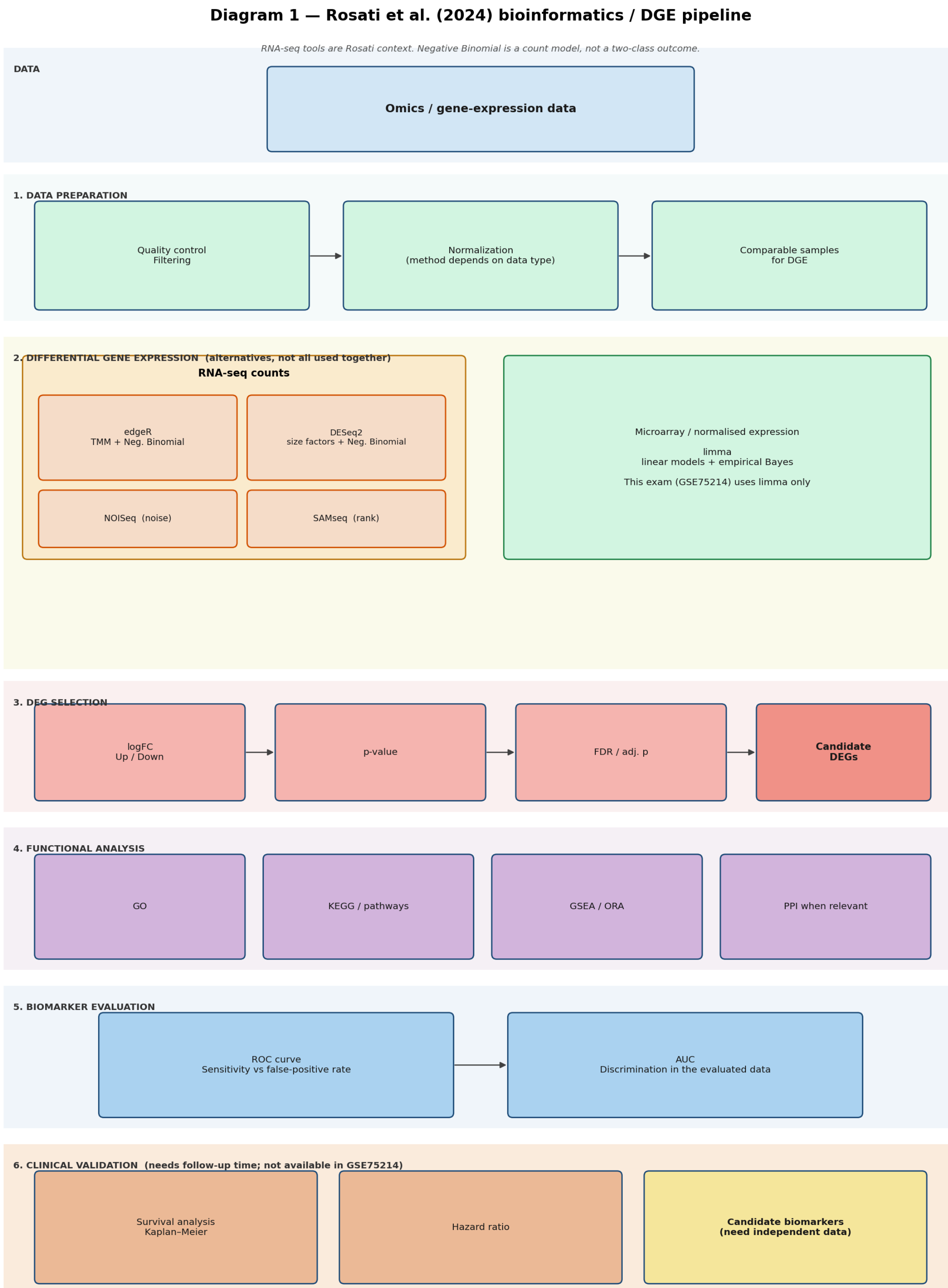


Diagram 1 moves from normalization to a differential expression model, to the DEG cutoffs, to visualization, to pathway enrichment, and ends with ROC and survival. Each box maps to a section of the Rosati review, so it is the map for Task 2.1 and Task 2.2.

In [7]:
```
show_fig(FIG / "diagram2_midterm1_ml_pipeline.png", width=620)
```

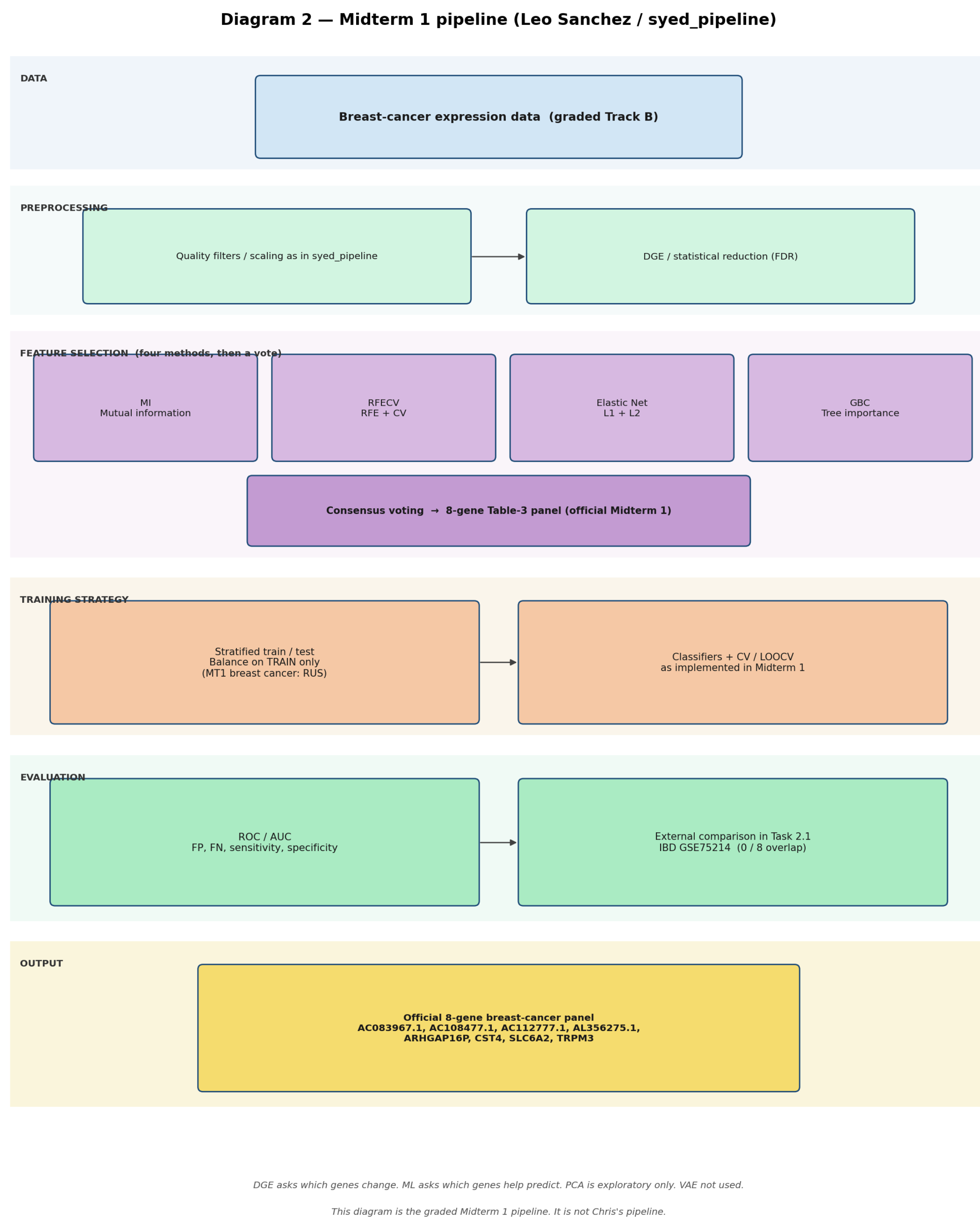


Diagram 2 runs on the Midterm 1 breast cancer set, with GSE75214 as the Question 3 validation cohort. It selects genes by their contribution to a classifier, which is the opposite of the significance test in Diagram 1. Task 2.1 runs both ideas on the same IBD data so the two can be compared fairly.

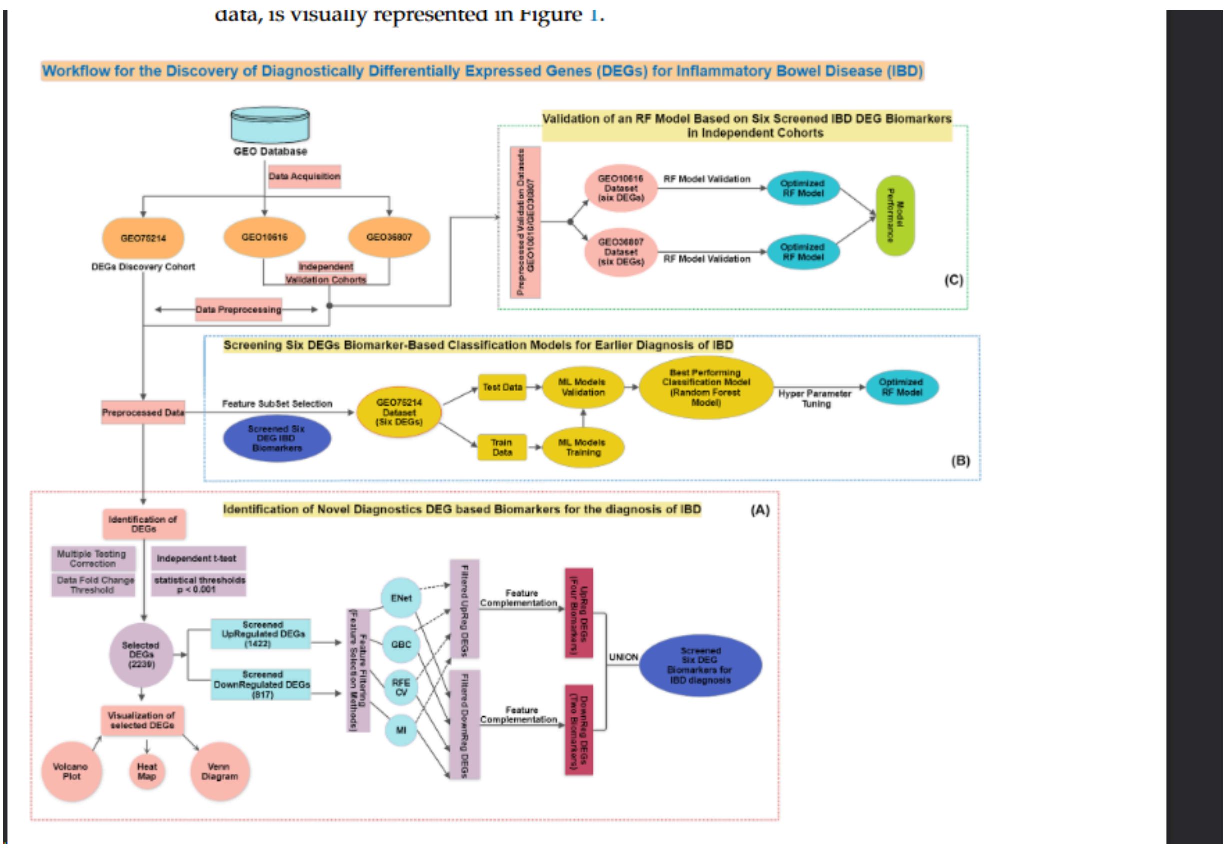
Syed et al. (2024) — reference workflow (Midterm 1 / Diagram 2)

The figure below is the **published Syed IBD biomarker workflow** (GEO75214 discovery cohort; GEO10616 and GEO36807 validation). Midterm 1 replicated this as `syed_pipeline`:

DEG filtering, four feature-selection methods (MI, RFECV, elastic net, gradient boosting) combined by **union**, then classifier training and independent validation. Diagram 2 in this exam is the same ML idea applied to breast cancer with GSE75214 as external validation — not a duplicate of Rosati DGE.

Reference only — our executed IBD numbers (535 DEGs, 16-gene ML panel, LDA CV AUC 0.9948) come from the notebooks in this folder, not from copying Syed's six-gene panel.

In [8]:
```
show_fig(FIG / "syed_pipeline_reference.png", width=720)
```



Task 2.1 Biomarkers in GSE75214 (inflammatory bowel disease)

The dataset is 172 disease against 22 control, log2 microarray on the GPL6244 platform. We ran the Rosati differential expression pipeline, then the Midterm 1 machine learning pipeline with SMOTE balancing, then evaluated and compared them.

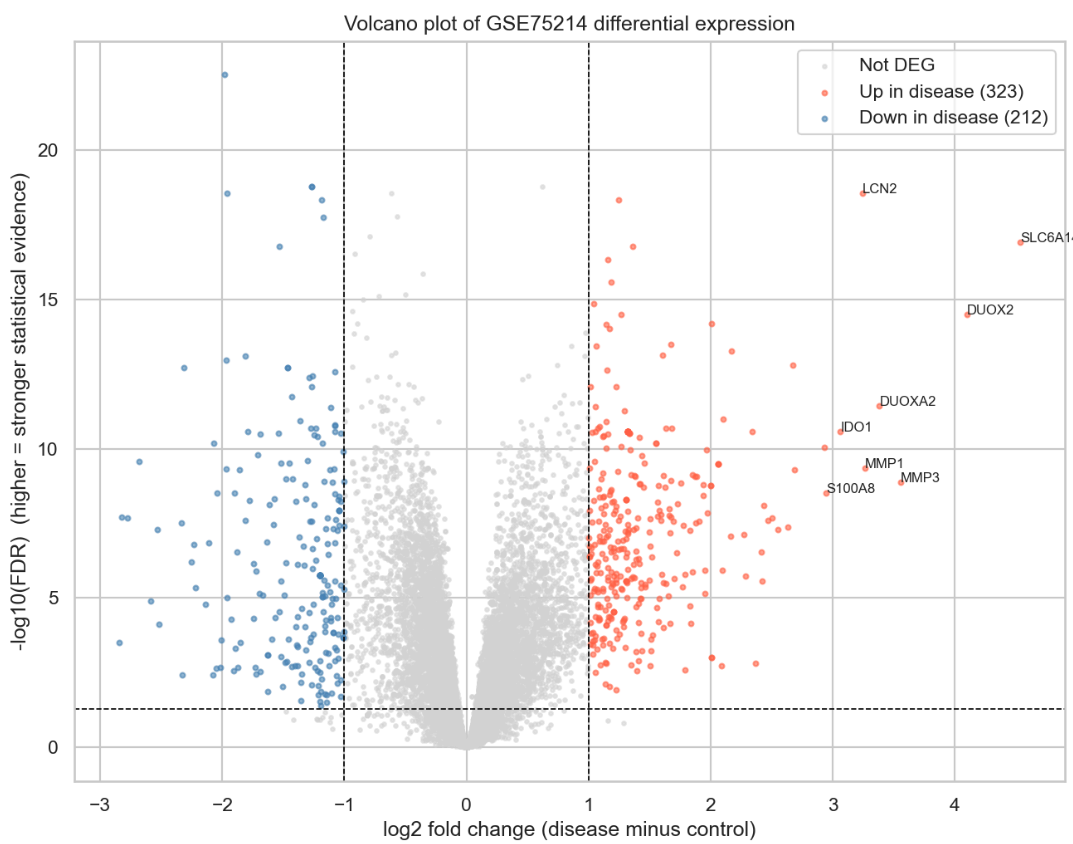
Differential expression result

In [9]:
```
dge1 = pd.read_csv(OUT / "dge_table_2_1.csv", index_col=0)
n_deg = int(dge1["DEG"].sum())
n_up = int((dge1["DEG"] & (dge1["direction"] == "up")).sum())
n_down = int((dge1["DEG"] & (dge1["direction"] == "down")).sum())
print(f"DEGs at FDR < 0.05 and |log2FC| >= 1.0: {n_deg}  (up {n_up}, down {n_down})
```

```
top = dge1[dge1["DEG"]].reindex(dge1[dge1["DEG"]]["log2FC"].abs().sort_values(ascen
display(top[["log2FC", "FDR", "direction"]].head(8).round(4))
show_fig(FIG / "task2_1_volcano.png", width=560)
```

```
DEGs at FDR < 0.05 and |log2FC| >= 1.0: 535  (up 323, down 212)
```

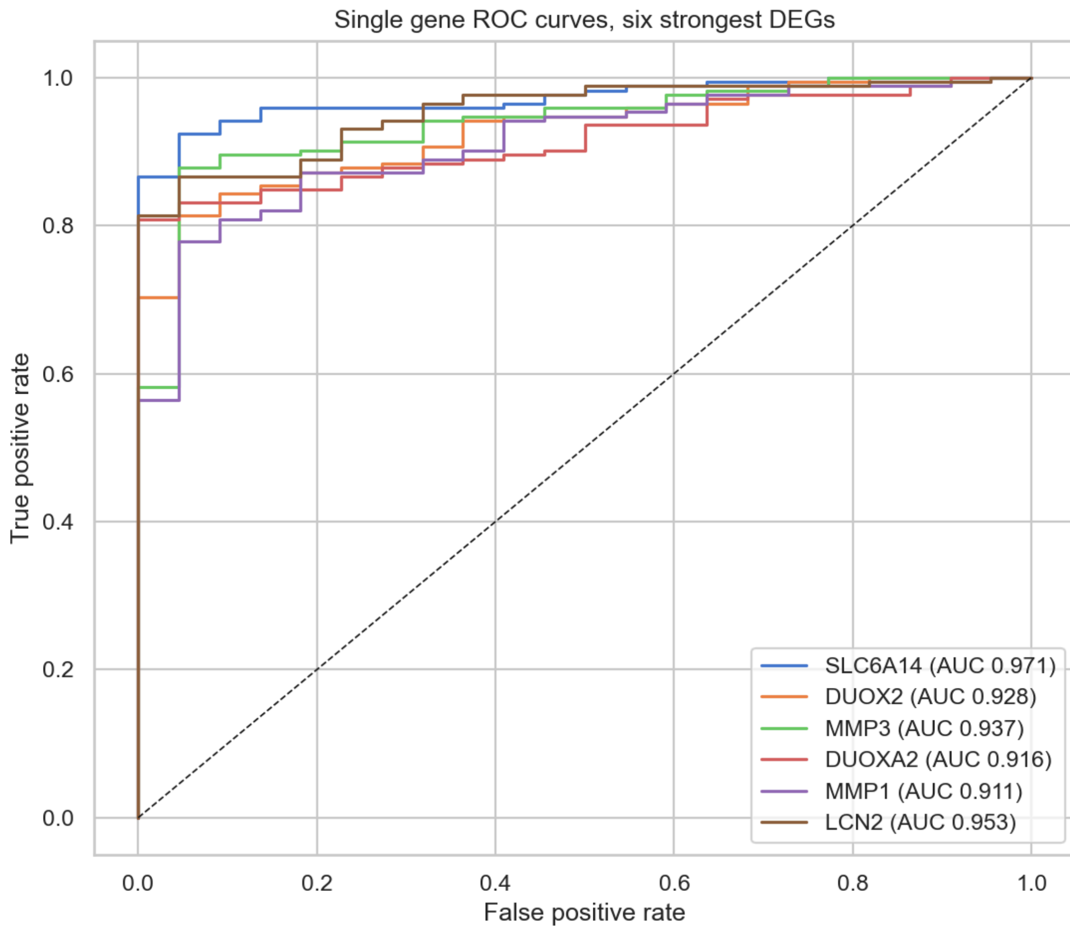
gene	log2FC	FDR	direction
SLC6A14	4.5331	0.0	up
DUOX2	4.0973	0.0	up
MMP3	3.5576	0.0	up
DUOXA2	3.3827	0.0	up
MMP1	3.2628	0.0	up
LCN2	3.2448	0.0	up
IDO1	3.0622	0.0	up
S100A8	2.9457	0.0	up



The pipeline finds 535 DEGs, and the strongest are named inflammatory bowel disease genes such as SLC6A14, DUOX2, LCN2, and IL1B. Seeing the correct disease genes at the top is the best sign the statistics are working, because these are the same markers the clinical literature reports for IBD.

Single gene ROC (Rosati section 4.1)

In [10]:
```
show_fig(FIG / "task2_1_roc_single.png", width=560)
```



Several strong DEGs also have high single-gene AUC in this dataset. That shows discriminatory potential here. It is not a clinical validation.

Machine learning panel and the balance issue

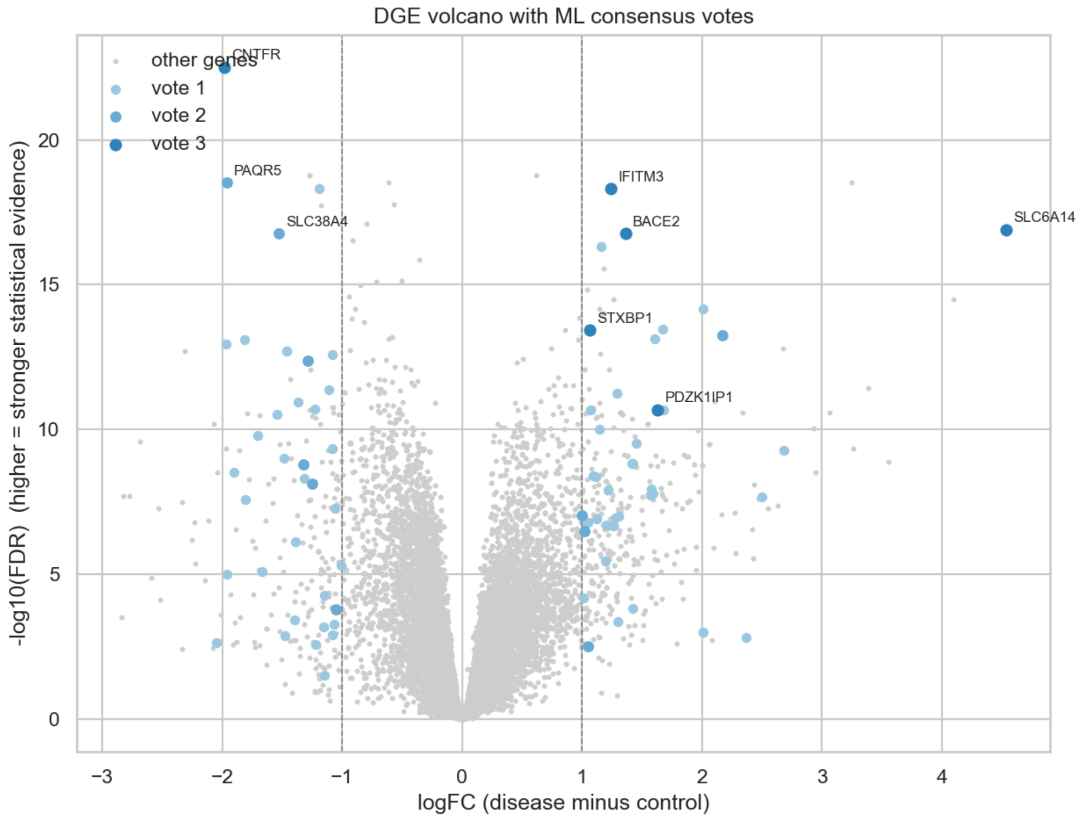
In [11]:
```
ml1 = pd.read_csv(OUT / "ml_biomarkers_2_1.csv", index_col=0)
print("Machine learning panel size:", len(ml1))
print("Genes chosen by 3 or more methods:", int((ml1["n_methods"] >= 3).sum()))
display(ml1.round(3))
```

```
Machine learning panel size: 16
Genes chosen by 3 or more methods: 6
```

gene	log2FC	FDR	direction	n_methods	MI	RFECV	ElasticNet	GBC
IFITM3	1.243	0.000	up	3	True	False	True	True
CNTFR	-1.984	0.000	down	3	True	False	True	True
PDZK1IP1	1.631	0.000	up	3	True	False	True	True
BACE2	1.364	0.000	up	3	True	True	True	False
SLC6A14	4.533	0.000	up	3	True	False	True	True
STXBP1	1.062	0.000	up	3	False	True	True	True
PAQR5	-1.960	0.000	down	2	True	False	False	True
SLC38A4	-1.531	0.000	down	2	True	False	True	False
TIMP1	2.173	0.000	up	2	True	False	False	True
NAAA	-1.289	0.000	down	2	True	False	False	True
TINAG	-1.325	0.000	down	2	False	True	True	False
ZG16B	1.024	0.000	up	2	False	True	True	False
SLC9A3	-1.053	0.000	down	2	False	True	False	True
CD34	1.001	0.000	up	2	False	True	True	False
APOBEC3B	-1.247	0.000	down	2	False	True	False	True
LOC101060256	1.053	0.003	up	2	False	False	True	True

The next figure puts ML votes on the DGE volcano. Grey = no TOPN vote. Darker blue = more methods.

In [12]:
```
volc = FIG / "task2_1_ml_consensus_volcano.png"
if volc.exists():
    show_fig(volc, width=560)
else:
    print("Run Notebook 2 to create the ML consensus volcano.")
```

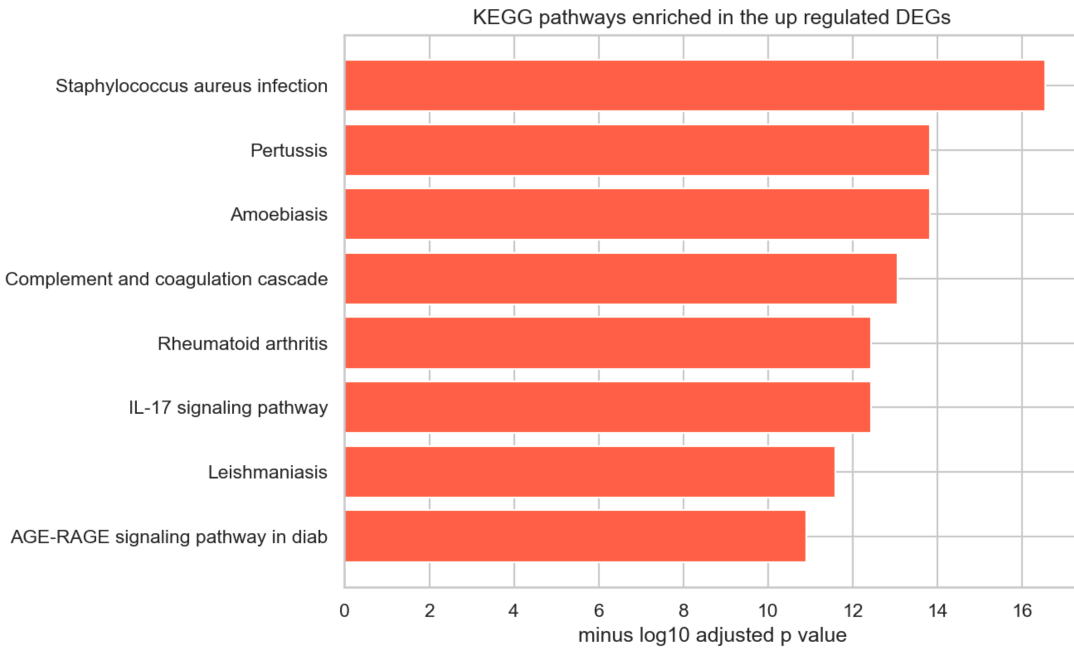


The four-method vote gives a compact ML panel. The size is printed from `ml_biomarkers_2_1.csv` / `report_data.json` (do not use an old 36-gene figure). SMOTE was used only in training. Zero overlap with the 8 Midterm 1 genes is expected: that panel is from breast cancer, not IBD.

Enrichment and the panel comparison

In [13]:

```
show_fig(FIG / "task2_1_enrichment.png", width=560)
cmp1 = pd.read_csv(OUT / "panel_comparison_2_1.csv")
display(cmp1)
```



	comparison	shared	out_of
0	Machine learning within statistics panel	6	16
1	Machine learning within Midterm 1 panel	0	16
2	Statistics within Midterm 1 panel	0	8
3	Midterm 1 genes present in GSE75214	3	8

Enrichment maps the up genes onto IL-17 signalling and other immune pathways, which is the core biology of IBD. The comparison table is the key result of Task 2.1. The statistics panel and the machine learning panel share 15 genes, so the two methods agree, but the Midterm 1 breast cancer panel does not transfer, since none of its 8 genes is an IBD DEG. A biomarker panel is specific to its disease.

Top 15 UP / Top 15 DOWN (GSE75214 + cervical)

Full code: `2_1_01_data_and_dge.ipynb` and `2_2_02_validate_pipelines.ipynb`. Genes in the ML panel are flagged in `in_ML_panel`.

In [14]:
```
import matplotlib.pyplot as plt
import numpy as np

dge1 = pd.read_csv(OUT / "dge_table_2_1.csv", index_col=0)
dge2 = pd.read_csv(OUT / "cervical_dge_table.csv", index_col=0)
deg_mask1 = dge1["DEG"] == True
deg_mask2 = (dge2["FDR"] < 0.05) & (dge2["log2FC"].abs() >= 1.5)

def top15_tables(dge, deg_mask, ml_genes):
    deg = dge[deg_mask].copy()
```

```
    up = deg[deg["direction"] == "up"].sort_values("log2FC", ascending=False).head(
    down = deg[deg["direction"] == "down"].sort_values("log2FC", ascending=True).he
    ml_set = set(ml_genes)
    def mark(df):
        out = df[["log2FC", "FDR"]].copy()
        out.index.name = "gene"
        out["in_ML_panel"] = [g in ml_set for g in out.index]
        return out.round(4)
    return mark(up), mark(down)

ml1 = pd.read_csv(OUT / "ml_biomarkers_2_1.csv")["gene"].tolist()
ml2 = pd.read_csv(OUT / "cervical_ml_panel.csv")["gene"].tolist()
up21, down21 = top15_tables(dge1, deg_mask1, ml1)
up22, down22 = top15_tables(dge2, deg_mask2, ml2)

print("GSE75214 — Top 15 UP (DEG, FDR<0.05, |log2FC|≥1)  |  genes in ML panel highl
display(up21)
print("GSE75214 — Top 15 DOWN")
display(down21)
print("Cervical merged — Top 15 UP")
display(up22)
print("Cervical merged — Top 15 DOWN")
display(down22)

overlap_up = sorted(set(up21.index) & set(up22.index))
overlap_down = sorted(set(down21.index) & set(down22.index))
print(f"Genes in top-15 UP of both tasks: {overlap_up}")
print(f"Genes in top-15 DOWN of both tasks: {overlap_down}")
```

```
GSE75214 — Top 15 UP (DEG, FDR<0.05, |log2FC|≥1)  |  genes in ML panel highlighted
```

gene	log2FC	FDR	in_ML_panel
SLC6A14	4.5331	0.0	True
DUOX2	4.0973	0.0	False
MMP3	3.5576	0.0	False
DUOXA2	3.3827	0.0	False
MMP1	3.2628	0.0	False
LCN2	3.2448	0.0	False
IDO1	3.0622	0.0	False
S100A8	2.9457	0.0	False
IL1B	2.9344	0.0	False
SAA2	2.6842	0.0	False
NOS2	2.6753	0.0	False
CHI3L1	2.6334	0.0	False
AQP9	2.5480	0.0	False
TCN1	2.5018	0.0	False
MMP12	2.4684	0.0	False

GSE75214 — Top 15 DOWN

gene	log2FC	FDR	in_ML_panel
MEP1B	-2.8415	0.0003	False
TMIGD1	-2.8216	0.0000	False
ABCG2	-2.7745	0.0000	False
CYP2B6	-2.6850	0.0000	False
UGT2A3	-2.5901	0.0000	False
HMGCS2	-2.5320	0.0000	False
SLC13A1	-2.5163	0.0001	False
SLC26A2	-2.3353	0.0000	False
CYP3A4	-2.3296	0.0038	False
MT1M	-2.3117	0.0000	False
PRKG2	-2.2525	0.0000	False
GUCA2A	-2.2303	0.0000	False
PCK1	-2.2202	0.0000	False
AQP8	-2.1405	0.0000	False
ABCB1	-2.1104	0.0000	False

```
Cervical merged — Top 15 UP
```

gene	log2FC	FDR	in_ML_panel
SPP1	3.7418	0.0	False
SYCP2	3.5861	0.0	True
CDKN2A	3.5719	0.0	True
CXCL8	3.5600	0.0	True
MMP12	3.3757	0.0	False
MMP1	3.2154	0.0	False
INHBA	3.0640	0.0	True
IFI44L	2.8616	0.0	False
AIM2	2.8498	0.0	True
NEK2	2.7310	0.0	False
ECT2	2.5564	0.0	False
APOC1	2.5443	0.0	False
CXCL13	2.4872	0.0	False
CA9	2.4590	0.0	True
CEP55	2.4144	0.0	False

```
Cervical merged — Top 15 DOWN
```

	log2FC	FDR	in_ML_panel
gene			
CRNN	-7.0438	0.0	True
CRISP3	-5.3675	0.0	False
MAL	-5.3654	0.0	False
KRT1	-5.3060	0.0	False
SLURP1	-4.7087	0.0	False
CRCT1	-4.5625	0.0	False
ENDOU	-4.5186	0.0	False
ALOX12	-4.4149	0.0	False
UPK1A	-4.3903	0.0	False
SPINK5	-4.3766	0.0	False
KRT4	-4.2514	0.0	False
PPP1R3C	-4.1164	0.0	True
KLK12	-3.9057	0.0	False
EDN3	-3.8824	0.0	False
TGM3	-3.8548	0.0	False

```
Genes in top-15 UP of both tasks: ['MMP1', 'MMP12']
Genes in top-15 DOWN of both tasks: []
```

Heatmap modules (Task 2.1)

In [15]:

```
mods = pd.read_csv(OUT / "task2_1_heatmap_gene_modules.csv")
print("Heatmap gene modules (from 2_1_01 clustering):")
for module_id, grp in mods.groupby("module"):
    genes = grp["gene"].tolist()
    print(f"Module {module_id} ({len(genes)} genes): {', '.join(genes)}")
display(mods)
```

```
Heatmap gene modules (from 2_1_01 clustering):
Module 1 (11 genes): SLC13A1, PRKG2, MEP1B, CYP3A4, CYP2B6, UGT2A3, ABCG2, ABCB1, PC
K1, TMIGD1, GUCA2A
Module 2 (4 genes): MT1M, AQP8, HMGCS2, SLC26A2
Module 3 (24 genes): IDO1, MMP12, CHI3L1, TIMP1, CXCL1, KYNU, MMP3, MMP1, PTGS2, S10
0A8, AQP9, IL1B, CXCL8, MUC1, TNIP3, MMP7, TCN1, VSIG1, NOS2, SAA2, SLC6A14, LCN2, D
UOX2, DUOXA2
Module 4 (1 genes): REG1B
```

	gene	cluster_position	module	log2FC	FDR	direction
0	SLC13A1	1	1	-2.516310	7.772739e-05	down
1	PRKG2	2	1	-2.252457	6.392515e-07	down
2	MEP1B	3	1	-2.841531	2.998035e-04	down
3	CYP3A4	4	1	-2.329615	3.845510e-03	down
4	CYP2B6	5	1	-2.685003	2.695643e-10	down
5	UGT2A3	6	1	-2.590060	1.280849e-05	down
6	ABCG2	7	1	-2.774521	2.028751e-08	down
7	ABCB1	8	1	-2.110422	1.397253e-07	down
8	PCK1	9	1	-2.220152	4.402362e-06	down
9	TMIGD1	10	1	-2.821645	1.931112e-08	down
10	GUCA2A	11	1	-2.230260	1.587601e-07	down
11	MT1M	12	2	-2.311677	1.950884e-13	down
12	AQP8	13	2	-2.140499	1.601429e-05	down
13	HMGCS2	14	2	-2.531992	5.227199e-08	down
14	SLC26A2	15	2	-2.335298	3.159881e-08	down
15	REG1B	16	4	2.366839	1.568446e-03	up
16	IDO1	17	3	3.062183	2.662948e-11	up
17	MMP12	18	3	2.468351	2.470499e-08	up
18	CHI3L1	19	3	2.633445	4.223694e-08	up
19	TIMP1	20	3	2.172807	5.405476e-14	up
20	CXCL1	21	3	2.338648	2.662948e-11	up
21	KYNU	22	3	2.166368	8.447880e-08	up
22	MMP3	23	3	3.557634	1.327190e-09	up
23	MMP1	24	3	3.262847	4.586332e-10	up
24	PTGS2	25	3	2.434932	8.117156e-09	up
25	S100A8	26	3	2.945704	3.087215e-09	up
26	AQP9	27	3	2.548022	5.224531e-08	up
27	IL1B	28	3	2.934362	8.998451e-11	up
28	CXCL8	29	3	2.275014	7.392575e-08	up
29	MUC1	30	3	2.103659	1.036297e-11	up

	gene	cluster_position	module	log2FC	FDR	direction
30	TNIP3	31	3	2.424859	2.747683e-06	up
31	MMP7	32	3	2.283376	1.852884e-06	up
32	TCN1	33	3	2.501841	2.068813e-08	up
33	VSIG1	34	3	2.417745	2.906218e-07	up
34	NOS2	35	3	2.675261	1.585763e-13	up
35	SAA2	36	3	2.684159	5.262321e-10	up
36	SLC6A14	37	3	4.533084	1.234089e-17	up
37	LCN2	38	3	3.244838	2.846337e-19	up
38	DUOX2	39	3	4.097317	3.213970e-15	up
39	DUOXA2	40	3	3.382728	3.634830e-12	up

Model comparison summary (Task 2.1)

One table combining **5-fold CV ROC-AUC** (LDA, logistic regression, random forest on both panels) with **80/20 hold-out** metrics (Accuracy, Precision, Recall, F1) for LDA/LR/RF and WELM/BWELM on the ML panel (same train/test split as feature selection).

CV rows report ROC-AUC only; hold-out rows include Precision (e.g. LDA shrinkage baseline). The LDA teaching plot (LD1 boxplot) is a visual check on the 16-gene ML panel — numbers come from this table. Built in `2_1_03_evaluation_and_comparison.ipynb` → `model_comparison_summary_2_1.csv`.

In [16]:
```
summary_path = OUT / "model_comparison_summary_2_1.csv"
if not summary_path.exists():
    print("Run 2_1_03_evaluation_and_comparison.ipynb to generate model_comparison_
else:
    unified = pd.read_csv(summary_path)
    print("Task 2.1 model comparison (CV + hold-out):")
    display(unified.round(4))
```

```
Task 2.1 model comparison (CV + hold-out):
```

	panel	model	evaluation	ROC-AUC	SD	Accuracy	Precision	Recall	F1	
0	Statistics panel (158)	LDA (linear baseline)	5-fold CV (SMOTE inside folds)	0.9821	0.0323	NaN	NaN	NaN	NaN	b sh
1	Statistics panel (158)	Logistic Regression	5-fold CV (SMOTE inside folds)	0.9942	0.0071	NaN	NaN	NaN	NaN	
2	Statistics panel (158)	Random Forest	5-fold CV (SMOTE inside folds)	0.9832	0.0086	NaN	NaN	NaN	NaN	tree
3	Machine learning panel (16)	LDA (linear baseline)	5-fold CV (SMOTE inside folds)	0.9948	0.0052	NaN	NaN	NaN	NaN	b sh
4	Machine learning panel (16)	Logistic Regression	5-fold CV (SMOTE inside folds)	0.9948	0.0064	NaN	NaN	NaN	NaN	
5	Machine learning panel (16)	Random Forest	5-fold CV (SMOTE inside folds)	0.9907	0.0077	NaN	NaN	NaN	NaN	tree
6	Machine learning panel (16)	WELM	80/20 hold-out	0.9857	NaN	0.8974	1.0	0.8857	0.9394	imb awa (tea
7	Machine learning panel (16)	BWELM	80/20 hold-out	0.9357	NaN	0.8974	1.0	0.8857	0.9394	imb awa (tea
8	Machine learning panel (16)	LDA (linear baseline)	80/20 hold-out (SMOTE on train)	0.9714	NaN	0.8974	1.0	0.8857	0.9394	sar as b sh
9	Machine learning panel (16)	Logistic Regression	80/20 hold-out (SMOTE on train)	0.9857	NaN	0.9487	1.0	0.9429	0.9706	sar as b sh

	panel	model	evaluation	ROC-AUC	SD	Accuracy	Precision	Recall	F1	
10	Machine learning panel (16)	Random Forest	80/20 hold-out (SMOTE on train)	1.0000	NaN	0.9744	1.0	0.9714	0.9855	sa as t sh

Task 2.2 Four merged cervical cancer datasets

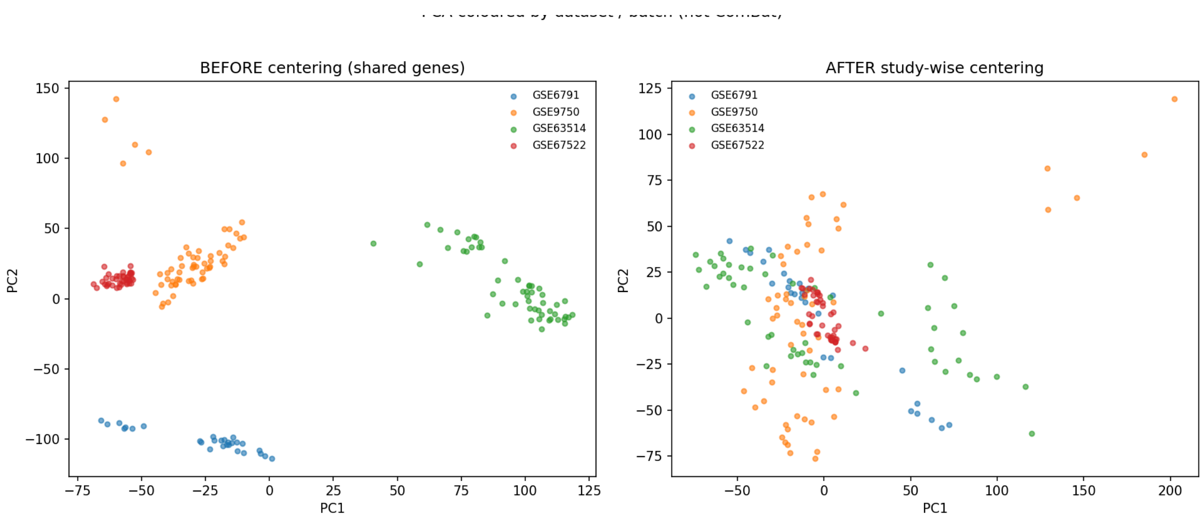
We merged GSE6791, GSE9750, GSE63514, and GSE67522 across three microarray platforms into one master set of 179 samples and the shared genes counted in the merge notebook, then validated both pipelines against Wang et al. (2022) at the cutoff |logFC| >= 1.5 and adjusted p below 0.05.

The merge

In [17]:

```
master = pd.read_csv(DATA / "cervical_master.csv")
print(
    "Master matrix:",
    master.shape[0],
    "samples ×",
    f"{master.shape[1] - 2:,}",
    "genes",
)
print("Cancer:", int((master["target"] == 1).sum()), " Normal:", int((master["targe
print("By dataset:")
print(master.groupby("batch")["target"].agg(["count", "sum"]).rename(columns={"sum"
show_fig(FIG / "task2_2_merge_pca.png", width=760)
```

```
Master matrix: 179 samples × 12,191 genes
Cancer: 101  Normal: 78
By dataset:
          count  cancer
batch
GSE63514     52      28
GSE67522     42      20
GSE6791      28      20
GSE9750      57      33
```



In the left PCA the four datasets overlap, which means the per dataset centering removed most of the platform batch effect. In the right PCA cancer and normal fall on opposite sides, so the biology survived the merge. This is the check that makes the merged analysis trustworthy.

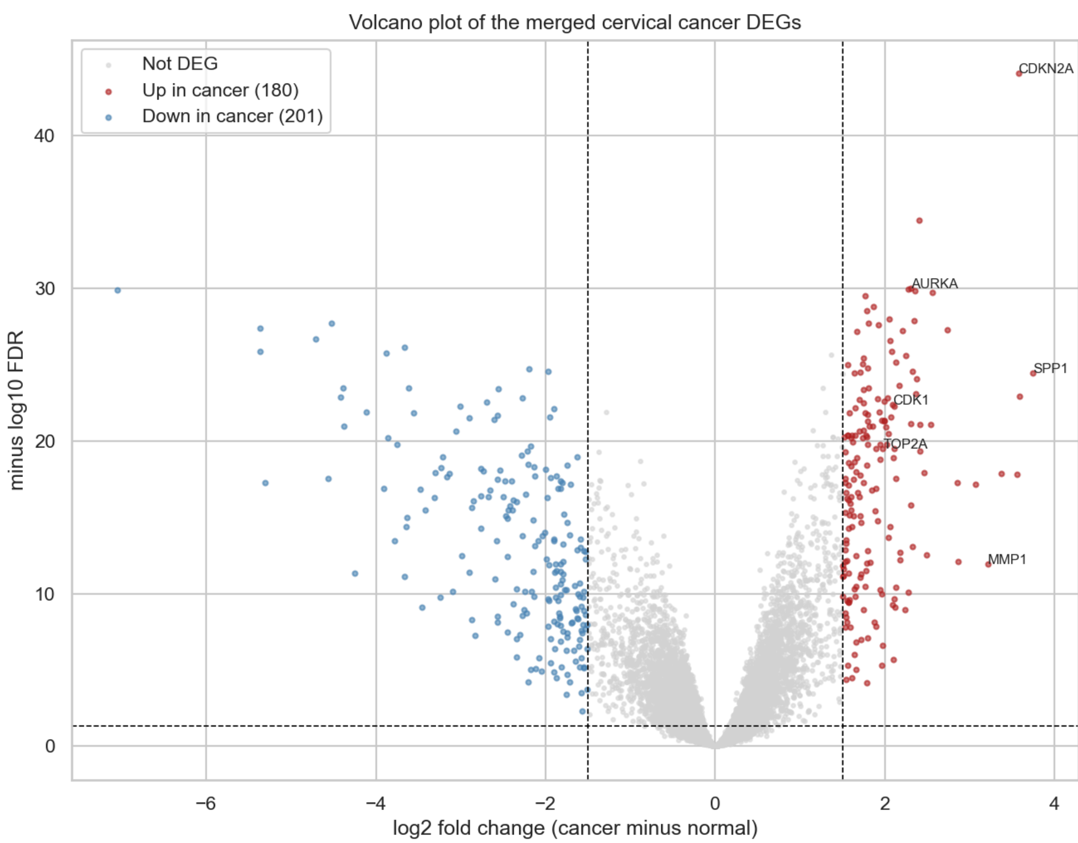
Differential expression and the Wang comparison

In [18]:

```
dge2 = pd.read_csv(OUT / "cervical_dge_table.csv", index_col=0)
print("Cervical DEGs at |logFC| >= 1.5 and adjP < 0.05:", int(dge2["DEG"].sum()))
wang_hub = ["CDK1", "CDC20", "AURKA", "TOP2A", "ASPM", "NCAPG", "KIF23", "CENPF", "
recovered = [g for g in wang_hub if g in dge2.index and dge2.loc[g, "DEG"]]
print("Wang hub genes recovered as DEGs:", len(recovered), "of 10")
display(dge2.reindex(wang_hub)[["log2FC", "FDR", "direction", "DEG"]].round(3))
show_fig(FIG / "task2_2_volcano.png", width=560)
```

```
Cervical DEGs at |logFC| >= 1.5 and adjP < 0.05: 381
Wang hub genes recovered as DEGs: 10 of 10
```

gene	log2FC	FDR	direction	DEG
CDK1	2.093	0.0	up	True
CDC20	1.845	0.0	up	True
AURKA	2.303	0.0	up	True
TOP2A	1.977	0.0	up	True
ASPM	2.307	0.0	up	True
NCAPG	1.666	0.0	up	True
KIF23	2.110	0.0	up	True
CENPF	1.993	0.0	up	True
KIF20A	2.019	0.0	up	True
PRC1	2.080	0.0	up	True



The merge gives 381 DEGs led by CDKN2A, the p16 marker of HPV driven cervical cancer. All ten hub genes that Wang et al. report are recovered as DEGs, every one upregulated in cancer. Recovering the whole published hub set from a differently sized merge is strong external validation of the pipeline.

Machine learning panel on the merged set

In [19]:
```
ml2 = pd.read_csv(OUT / "cervical_ml_panel.csv", index_col=0)
print("Machine learning panel size:", len(ml2))
print("AURKA in the panel:", "AURKA" in ml2.index)
display(ml2.round(3))
```

```
Machine learning panel size: 47
AURKA in the panel: True
```

gene	log2FC	FDR	direction	n_methods
SMC4	1.921	0.0	up	4
THSD4	-2.690	0.0	down	4
CDKN2A	3.572	0.0	up	4
UBE2C	1.744	0.0	up	4
TACC3	1.744	0.0	up	4
TYMS	2.053	0.0	up	3
TIMELESS	1.784	0.0	up	3
RAD54L	2.207	0.0	up	3
MCM4	1.641	0.0	up	3
FANCI	1.804	0.0	up	3
PIGR	-2.075	0.0	down	3
TBX3	-2.125	0.0	down	3
TUBB2A	-1.942	0.0	down	3
PHYHIP	-2.278	0.0	down	2
MELK	2.354	0.0	up	2
BBOX1	-3.611	0.0	down	2
PRC1	2.080	0.0	up	2
CRNN	-7.044	0.0	down	2
STIL	1.794	0.0	up	2
AURKA	2.303	0.0	up	2
CDC25B	1.820	0.0	up	2
CA9	2.459	0.0	up	2
KNTC1	2.407	0.0	up	2
SULF1	1.719	0.0	up	2
CELSR3	1.996	0.0	up	2
PPP1R3C	-4.116	0.0	down	2
SLC15A3	1.708	0.0	up	2
KIFC1	1.775	0.0	up	2
AIM2	2.850	0.0	up	2

gene	log2FC	FDR	direction	n_methods
MUC5B	-1.937	0.0	down	2
LHX2	1.945	0.0	up	2
SPINK2	-1.820	0.0	down	2
CYP3A5	-2.123	0.0	down	2
TFF3	-2.051	0.0	down	2
MASP1	-1.595	0.0	down	2
LGALSL	-2.013	0.0	down	2
CRISP2	-2.481	0.0	down	2
ZSCAN18	-2.001	0.0	down	2
ZNF415	-1.747	0.0	down	2
INHBA	3.064	0.0	up	2
CXCR2	-2.134	0.0	down	2
CXCL8	3.560	0.0	up	2
DPP4	-1.824	0.0	down	2
LPAR6	-1.593	0.0	down	2
DUSP1	-1.548	0.0	down	2
KCNK7	-2.180	0.0	down	2
SYCP2	3.586	0.0	up	2

The machine learning pipeline cuts the 381 DEGs to a 47 gene panel that reaches a cross validated AUC near 0.997. The panel keeps AURKA, the single gene Wang et al. tie to worse survival, so the data driven method lands on their most important marker on its own.

Oral defense (consolidated)

Data / EDA

- What are the dimensions of GSE75214 (samples × probes, then samples × genes)?
- How many disease and control samples? Missing values?
- What are the dimensions of each cervical dataset before merge?
- Are rows samples or probes in raw GEO files?

DGE

- What is logFC? Why separate UP and DOWN?
- What is FDR, and why not raw p-values only?
- Why limma instead of DESeq2 or edgeR for GSE75214?
- What does the volcano show? What can we not conclude?
- Which DEG cutoffs does the exam require?

Feature selection

- What does MI measure? How does RFECV work (support vs ranking)?
- Why TOPN consensus? Why several FS methods?
- Where is SMOTE applied, and why not before the split?

Evaluation

- FP vs FN? Sensitivity vs specificity?
- What does AUC mean? How is FDR different from classifier FP/FN?
- What is LDA, and why use it as a baseline?
- Why can AUC near 0.99 still be optimistic?

Merge / integration

- Why did gene count decrease after merge?
- Why do sample counts differ from Wang?
- Study-wise centering vs ComBat — what did we run?
- Why is Wang 10/10 not an exact replication?
- Why is Midterm 1 overlap zero (not a failed midterm)?

Conclusion

The Rosati-style DGE step finds genes with a statistically important expression change. The Syed / Midterm 1 style feature-selection step finds genes that help predict the class. Together they give a shorter candidate list. High AUC on GSE75214 can still be optimistic because selection and evaluation use the same cohort. The cervical analysis recovers 10/10 Wang hub genes as external agreement, not as an exact copy of Wang's pipeline.